Physics 827: Homework Set No. 4

Due date: Friday, October 21, 2011, 1:00pm in PRB M2043 (Biao Huang's office)

Total point value of set: 60 points

Problem 1 (10 pts.): Show that

(i) $\displaystyle\int_{-\infty}^{\infty} dx\, e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}}$, (ii) $\displaystyle\int_{0}^{\infty} dx\, x\, e^{-\alpha x^2} = \frac{1}{2\alpha}$,

(iii) $\displaystyle\int_{-\infty}^{\infty} dx\, x^2\, e^{-\alpha x^2} = \frac{1}{2\alpha}\sqrt{\frac{\pi}{\alpha}}$, (iv) $\displaystyle\int_{-\infty}^{\infty} dx\, e^{-\alpha x^2+\beta x} = e^{\beta^2/(4\alpha)}\frac{1}{2\alpha}$.

You can't use Mathematica for this problem!

Problem 2 (10 pts.): Exercise 4.2.1 in Shankar (p.129). The answers are given in Shankar, but you will be graded on the correctness of each step in your work leading to the answers.

Problem 3 (10 pts.): If $[\hat{A},\hat{B}] \neq 0$, λ a real or complex scalar, show that

(i) $e^{-\lambda\hat{A}}\,\hat{B}^n\,e^{\lambda\hat{A}} = \left(e^{-\lambda\hat{A}}\,\hat{B}\,e^{\lambda\hat{A}}\right)^n$, (ii) $e^{-\lambda\hat{A}}\,\hat{F}(\hat{B})\,e^{\lambda\hat{A}} = \hat{F}(e^{-\lambda\hat{A}}\,\hat{B}\,e^{\lambda\hat{A}})$,

where n is integer and $\hat{F}(\hat{B})$ is an arbitrary function of $\hat{B}$ that does not depend on any other operators.

Problem 4 (20 pts.): If $[\hat{A},\hat{B}] = \hat{I}$ ($\hat{I}$ = identity operator), λ a scalar, show

(i) $e^{-\lambda\hat{B}\hat{A}}\,\hat{A}\,e^{\lambda\hat{B}\hat{A}} = e^{\lambda}\,\hat{A}$, (ii) $e^{-\lambda\hat{B}\hat{A}}\,\hat{B}\,e^{\lambda\hat{B}\hat{A}} = e^{-\lambda}\,\hat{B}$,

(iii) $e^{-(\lambda/2)(\hat{B}^2-\hat{A}^2)}\,\hat{A}\,e^{(\lambda/2)(\hat{B}^2-\hat{A}^2)} = \hat{A}\cosh\lambda + \hat{B}\sinh\lambda$,

(iv) $e^{-(\lambda/2)(\hat{B}^2-\hat{A}^2)}\,\hat{B}\,e^{(\lambda/2)(\hat{B}^2-\hat{A}^2)} = \hat{B}\cosh\lambda + \hat{A}\sinh\lambda$.

Problem 5 (10 pts.):

(a) Write down the x-components of the eigenvectors $|p\rangle$ of the momentum operator $\hat{P} = \hbar\hat{K}$.

(b) Compute the x-components of the Hilbert space vector obtained by adding all such vectors corresponding to eigenvalues between p and $p+\Delta p$ and dividing the result by $\sqrt{\Delta p}$.

(c) Divide the result (b) once more by $\sqrt{\Delta p}$ and take the limit $\Delta p \to 0$ — what do you get?

(d) What is the norm of the vector obtained in (b)?

(e) Taking two such vectors, one obtained as in (b), the other by performing an analogous "average" of $\hat{P}$-eigenvectors over the interval $[p', p'+\Delta p]$ where $|p-p'| = \Delta p$, compute their inner product.

(f) Give a physical interpretation of your result.

Problem 1

(i) Show $I(\alpha) \equiv \int_{-\infty}^{\infty} dx\, e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}}$

Trick: compute I^2 and use polar coordinates in x-y plane:

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dx\,dy\; e^{-\alpha(x^2+y^2)} = \int_0^{\infty} r\,dr \int_0^{2\pi} d\varphi\; e^{-\alpha r^2} = 2\pi \int_0^{\infty} \frac{d(r^2)}{2}\, e^{-\alpha r^2}$$

$$(t=\alpha r^2) = \frac{\pi}{\alpha}\int_0^{\infty} dt\, e^{-t} = \frac{\pi}{\alpha} \Rightarrow I = \sqrt{\frac{\pi}{\alpha}} \quad \diamond$$

(ii) $$\int_0^{\infty} dx\, x\, e^{-\alpha x^2} \overset{t=\alpha x^2}{=} \frac{1}{2\alpha}\int_0^{\infty} dt\, e^{-t} = \frac{1}{2\alpha} \quad \diamond$$

(iii) $$\int_{-\infty}^{\infty} dx\, x^2 e^{-\alpha x^2} = \left(-\frac{d}{d\alpha}\right)\int_{-\infty}^{\infty} dx\, e^{-\alpha x^2} = -\frac{d}{d\alpha} I(\alpha)$$

$$= \sqrt{\pi}\,\frac{1}{2}\alpha^{-3/2} = \frac{1}{2\alpha}\sqrt{\frac{\pi}{\alpha}} \quad \diamond$$

(iv) $$\int_{-\infty}^{\infty} dx\, e^{-\alpha x^2 + \beta x} = \int_{-\infty}^{\infty} e^{-\alpha\left(x-\frac{\beta}{2\alpha}\right)^2} e^{\frac{\beta^2}{4\alpha}}\, dx$$

complete the square

$$\overset{t=x-\frac{\beta}{2\alpha}}{=} e^{\beta^2/4\alpha} \underbrace{\int_{-\infty}^{\infty} dt\, e^{-\alpha t^2}}_{I(\alpha)} = e^{\beta^2/4\alpha}\sqrt{\frac{\pi}{\alpha}} \quad \diamond$$

Problem 2

(1) The only possible values one can obtain when measuring L_z are its eigenvalues.

To compute eigenvalues, solve characteristic equation:

$$\det(\hat{L}_z - \lambda \hat{I}) = 0 \quad \Rightarrow \quad \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & -1-\lambda \end{vmatrix} = \lambda(1-\lambda)(1+\lambda) = 0$$

$$\Rightarrow \boxed{\lambda_1 = 0 \ , \ \lambda_2 = 1 \ , \ \lambda_3 = -1}$$

(2) The eigenstate with eigenvalue $L_z = \lambda_2 = 1$ has the following components:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{matrix} a = a \\ 0 = b \\ -c = c \end{matrix} \Rightarrow |L_z = 1\rangle \equiv |1\rangle \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

(normalized)

$$\langle 1 | \hat{L}_x | 1 \rangle = (1,0,0)\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(1,0,0)\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$\langle 1 | \hat{L}_x^2 | 1 \rangle = (1,0,0)\frac{1}{2}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \frac{1}{2}(1,0,0)\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2}(1,0,0)\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2}$$

$$\Delta L_x = \sqrt{\langle \hat{L}_x \rangle^2 - \langle L_x \rangle^2} = \sqrt{\tfrac{1}{2} - 0} = \frac{1}{\sqrt{2}}.$$

(3) The other two eigenstates of $\hat{L}_z$ are:

$$\lambda_1 = 0: \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0 \Rightarrow \begin{matrix} a = 0 \\ b \text{ arbitrary} \\ c = 0 \end{matrix} \Rightarrow |L_z = 0\rangle \equiv |0\rangle \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\lambda_3 = -1: \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = -\begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{matrix} a = -a \\ 0 = b \\ -c = -c \end{matrix} \Rightarrow |L_z = -1\rangle \equiv |-1\rangle \leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

So $\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}$ are the components of $|1\rangle, |0\rangle, |-1\rangle$

in the basis in which $\hat{L}_x$, $\hat{L}_y$ and $\hat{L}_z$ have the given matrix elements.

Next we have to find the eigenvalues of $\hat{L}_x$:

$$\det(\hat{L}_x - \lambda\hat{I}) = 0 \Rightarrow \begin{vmatrix} -\lambda & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -\lambda & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda\left(\lambda^2 - \frac{1}{2}\right) + \frac{1}{\sqrt{2}}\left(+\frac{\lambda}{\sqrt{2}}\right) + 0 = 0 \Rightarrow -\lambda^3 + \frac{\lambda}{2} + \frac{\lambda}{2} = -\lambda^3 + \lambda$$

$$= \lambda(1-\lambda^2) = \lambda(1-\lambda)(1+\lambda) = 0$$

$$\Rightarrow \boxed{\lambda_1 = 0\,,\ \lambda_2 = 1\,,\ \lambda_3 = -1}$$ same eigenvalues as L_z!

In the basis where $\hat{L}_x$ has the given matrix elements, the components of the corresponding three eigenstates are:

$$L_x = 1:\ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \sqrt{2}\begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{matrix} b = \sqrt{2}a \\ a + c = \sqrt{2}b \\ b = \sqrt{2}c \end{matrix} \Rightarrow a = c = \frac{b}{\sqrt{2}}$$

$$\Rightarrow |L_x = 1\rangle \longleftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix} 1/\sqrt{2} \\ 1 \\ 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 1/\sqrt{2} \\ 1/2 \end{pmatrix}$$

$$L_x = 0:\ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0 \Rightarrow \begin{matrix} b = 0 \\ a + c = 0 \\ b = 0 \end{matrix} \Rightarrow |L_x = 0\rangle \leftrightarrow \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$$

$$L_x = -1:\ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = -\sqrt{2}\begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{matrix} b = -\sqrt{2}a \\ a + c = -\sqrt{2}b \\ b = -\sqrt{2}c \end{matrix} \Rightarrow a = c = -\frac{b}{\sqrt{2}}$$

$$\Rightarrow |L_x = -1\rangle \leftrightarrow \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix}$$

Next, we should compute the components of these 3 $\hat{L}_x$-eigenstates in the $\{|1\rangle, |0\rangle, |-1\rangle\}$-basis of $\hat{L}_z$-eigenstates. But since $\hat{L}_z$ is diagonal in the basis in which $\hat{L}_x$, $\hat{L}_y$ and $\hat{L}_z$ are given, the basis that Shankar used to write down the matrix elements of $\hat{L}_x, \hat{L}_y, \hat{L}_z$ is the $\hat{L}_z$-eigenbasis. So the components of $|L_x = 1, 0, -1\rangle$ in the given basis that we just calculated are their components in the $\hat{L}_z$-eigenbasis.

(4) If we measure $\hat{L}_x$ in any state, the possible outcomes are any one of the eigenvalues $L_x = \pm 1, 0$.

The probabilities for $L_x = \pm 1, 0$ in the state $|-1\rangle \equiv |L_z = -1\rangle$ are:

$$\underset{\sim}{P(L_x = 1)} = |\langle L_x = 1 | L_z = -1\rangle|^2 = \left[\left(\tfrac{1}{2}, \tfrac{1}{\sqrt{2}}, \tfrac{1}{2}\right)\begin{pmatrix}0\\0\\1\end{pmatrix}\right]^2 = \underset{\sim}{\tfrac{1}{4}}$$

$$\underset{\sim}{P(L_x = 0)} = |\langle L_x = 0 | L_z = -1\rangle|^2 = \left[\left(\tfrac{1}{\sqrt{2}}, 0, -\tfrac{1}{\sqrt{2}}\right)\begin{pmatrix}0\\0\\1\end{pmatrix}\right]^2 = \underset{\sim}{\tfrac{1}{2}}$$

$$\underset{\sim}{P(L_x = -1)} = |\langle L_x = -1 | L_z = -1\rangle|^2 = \left|\left(\tfrac{1}{2}, -\tfrac{1}{\sqrt{2}}, \tfrac{1}{2}\right)\begin{pmatrix}0\\0\\1\end{pmatrix}\right|^2 = \underset{\sim}{\tfrac{1}{4}}$$

(5). $|\psi\rangle \leftrightarrow \begin{pmatrix}1/2\\1/2\\1/\sqrt{2}\end{pmatrix}$ in L_z-basis

$\hat{L}_z^2$ measured, $L_z^2 = 1$ found $\Rightarrow L_z = \pm 1$

$$\Rightarrow \underset{\sim}{|\psi\rangle_{after}} = N\left(|1\rangle\langle 1| + |-1\rangle\langle -1|\right)|\psi\rangle$$ where N normalizes the state to 1

$$\leftrightarrow \frac{1}{\sqrt{1/4 + \frac{1}{2}}}\begin{pmatrix}1/2\\0\\1/\sqrt{2}\end{pmatrix} = \frac{2}{\sqrt{3}}\begin{pmatrix}1/2\\0\\1/\sqrt{2}\end{pmatrix} = \underset{\sim}{\begin{pmatrix}1/\sqrt{3}\\0\\\sqrt{2/3}\end{pmatrix}}$$

$$\underset{\sim}{P(L_z^2 = 1)} = P(L_z = 1 \text{ or } L_z = -1) = P(L_z = 1) + P(L_z = -1) = |\langle 1|\psi\rangle|^2 + |\langle -1|\psi\rangle|^2 = \tfrac{1}{2} + \tfrac{1}{4}$$

$$= \underset{\sim}{\tfrac{3}{4}}$$

If $\hat{L}_z$ is measured in the state $|\psi\rangle$, the possible outcomes are:

$$L_z = 1 \quad \text{with} \quad P_\psi(L_z=1) = |\langle 1|\psi\rangle|^2 = \left|(1,0,0)\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{pmatrix}\right|^2 = \frac{1}{4}$$

$$L_z = 0 \quad \text{with} \quad P_\psi(L_z=0) = |\langle 0|\psi\rangle|^2 = \frac{1}{4}$$

$$L_z = -1 \quad \text{with} \quad P_\psi(L_z=-1) = |\langle -1|\psi\rangle|^2 = \frac{1}{2}$$

If $\hat{L}_z$ is measured after $\hat{L}_z^2$ was measured and $L_z^2=1$ was found, then $\begin{cases} P'(L_z=1) = \frac{1}{3} \\ P'(L_z=0) = 0 \\ P'(L_z=-1) = \frac{2}{3} \end{cases}$

(6) A particle in a state with $P(L_z=1) = \frac{1}{4}$, $P(L_z=0) = \frac{1}{2}$, $P(L_z=-1) = \frac{1}{4}$

is in a state

$$|\psi\rangle = \frac{\alpha|1\rangle + \beta|0\rangle + \gamma|-1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2 + |\gamma|^2}}$$

with $\dfrac{|\alpha|^2}{|\alpha|^2+|\beta|^2+|\gamma|^2} = \dfrac{1}{4}$, $\dfrac{|\beta|^2}{|\alpha|^2+|\beta|^2+|\gamma|^2} = \dfrac{1}{2}$, $\dfrac{|\gamma|^2}{|\alpha|^2+|\beta|^2+|\gamma|^2} = \dfrac{1}{4}$.

$$\Rightarrow \boxed{\alpha = \frac{e^{i\delta_1}}{2}, \quad \beta = \frac{e^{i\delta_2}}{\sqrt{2}}, \quad \gamma = \frac{e^{i\delta_3}}{2}}$$ as stated.

Here $\delta_1, \delta_2, \delta_3$ are arbitrary real numbers.

The values of the phases matter when measuring an observable that is incompatible with $\hat{L}_z$, e.g. $\hat{L}_x$:

$$P_\psi(L_x=0) = \left|\left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)\begin{pmatrix} \frac{e^{i\delta_1}}{2} \\ \frac{e^{i\delta_2}}{\sqrt{2}} \\ \frac{e^{i\delta_3}}{2} \end{pmatrix}\right|^2 = \frac{1}{8}\left|e^{i\delta_1} - e^{i\delta_3}\right|^2$$

$$= \frac{1}{8}\left(e^{i\delta_1} - e^{i\delta_3}\right)\left(e^{-i\delta_1} - e^{-i\delta_3}\right) = \frac{1}{8}\left(1 - e^{i(\delta_1-\delta_3)} - e^{-i(\delta_1-\delta_3)} + 1\right)$$

$$= \frac{1}{2}\left(1 - \cos(\delta_1 - \delta_3)\right)$$

Depending on the phases, this can vary between 0 and 1 !

Problem 3

(i) $$e^{-\lambda\hat{A}}\hat{B}^n e^{\lambda\hat{A}} = \overbrace{\left(e^{-\lambda\hat{A}}\hat{B}\underbrace{e^{\lambda\hat{A}}\right)\left(e^{-\lambda\hat{A}}}_{\hat{I}}\hat{B}\underbrace{e^{+\lambda\hat{A}}\right)\cdots\left(e^{-\lambda\hat{A}}}_{\hat{I}}\hat{B}e^{\lambda\hat{A}}\right)}^{n \text{ factors}}$$

$$= \left(e^{-\lambda\hat{A}}\hat{B}e^{\lambda\hat{A}}\right)^n \quad \diamond$$

(ii) Functions of operators are defined through their power series $\hat{F}(\hat{B}) = \sum_{n=0}^{\infty} a_n \hat{B}^n$:

$$e^{-\lambda\hat{A}}\hat{F}(\hat{B})e^{\lambda\hat{A}} = \sum_{n=0}^{\infty} a_n e^{-\lambda\hat{A}}\hat{B}^n e^{\lambda\hat{A}} \overset{(i)}{=} \sum_{n=0}^{\infty} a_n \left(e^{-\lambda\hat{A}}\hat{B}e^{\lambda\hat{A}}\right)^n$$

$$= \hat{F}\left(e^{-\lambda\hat{A}}\hat{B}e^{\lambda\hat{A}}\right) \quad \diamond$$

Problem 4 $\quad [\hat{A},\hat{B}] = \hat{I}$

(i) $e^{-\lambda\hat{B}\hat{A}}\hat{A}e^{\lambda\hat{B}\hat{A}} \overset{\text{problem 9, HW2}}{=} \sum_{n=0}^{\infty}\frac{1}{n!}[-\lambda\hat{B}\hat{A},\hat{A}]_{(n)} = \sum_{n=0}^{\infty}\frac{(-\lambda)^n}{n!}[\hat{B}\hat{A},\hat{A}]_{(n)}$

$$[\hat{B}\hat{A},\hat{A}]_{(0)} = \hat{A}\ ;\quad [\hat{B}\hat{A},\hat{A}]_{(1)} = [\hat{B}\hat{A},\hat{A}] = \hat{B}\hat{A}^2 - \hat{A}\hat{B}\hat{A} = [\hat{B},\hat{A}]\hat{A} = -\hat{A}$$

$$[\hat{B}\hat{A},\hat{A}]_{(n)} = [\hat{B}\hat{A},[\hat{B}\hat{A},\hat{A}]_{(n-1)}] = [\hat{B}\hat{A},(-1)^{n-1}\hat{A}] = (-1)^{n-1}(\hat{B}\hat{A}^2 - \hat{A}\hat{B}\hat{A})$$
$$= (-1)^n\hat{A}$$

$$\Rightarrow \underline{e^{-\lambda\hat{B}\hat{A}}\hat{A}e^{\lambda\hat{B}\hat{A}}} = \sum_{n=0}^{\infty}\frac{(-\lambda)^n(-1)^n}{n!}\hat{A} = \left(\sum_{n=0}^{\infty}\frac{\lambda^n}{n!}\right)\hat{A} = \underline{e^{\lambda}\hat{A}}$$

(ii) Here we need $[\hat{B}\hat{A},\hat{B}]_{(n)}$: $[\hat{B}\hat{A},\hat{B}]_{(0)} = \hat{B}$; $[\hat{B}\hat{A},\hat{B}]_{(1)} = \hat{B}\hat{A}\hat{B} - \hat{B}^2A$
$= \hat{B}[\hat{A},\hat{B}] = \hat{B}$

$$[\hat{B}\hat{A},\hat{B}]_{(n)} = [\hat{B}\hat{A},[\hat{B}\hat{A},\hat{B}]_{(n-1)}] = [\hat{B}\hat{A},\hat{B}] = \hat{B}$$

$$\Rightarrow \underline{e^{-\lambda\hat{B}\hat{A}}\hat{B}e^{\lambda\hat{B}\hat{A}}} = \sum_{n=0}^{\infty}\frac{(-\lambda)^n}{n!}[\hat{B}\hat{A},\hat{B}]_{(n)} = \left(\sum_{n=0}^{\infty}\frac{(-\lambda)^n}{n!}\right)\hat{B} = \underline{e^{-\lambda}\hat{B}}$$

(iii) Here we need $[-\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{A}]_{(n)}$.

$$[-\tfrac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{A}]_{(0)} = \hat{A}$$

$$[-\tfrac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{A}]_{(1)} = -\frac{\lambda}{2}\left(\hat{B}^2\hat{A} - \hat{A}^3 - \hat{A}\hat{B}^2 + \hat{A}^3\right) = -\frac{\lambda}{2}\left(\hat{B}^2\hat{A} - \hat{B}\hat{A}\hat{B} + \hat{B}\hat{A}\hat{B} - \hat{A}\hat{B}^2\right)$$
$$= -\frac{\lambda}{2}\left(\hat{B}[\hat{B},\hat{A}] + [\hat{B},\hat{A}]\hat{B}\right) = +\lambda\hat{B}$$

$$[-\tfrac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{A}]_{(2)} = -\frac{\lambda}{2}[\hat{B}^2-\hat{A}^2,\lambda\hat{B}] = +\frac{\lambda^2}{2}[\hat{A}^2,\hat{B}] = \frac{\lambda^2}{2}(\hat{A}[\hat{A},\hat{B}] + [\hat{A},\hat{B}]\hat{A})$$
$$= \lambda^2 A$$

$\rightarrow$ we alternate between $\hat{A}$ and $\hat{B}$, picking up additional factor λ in each step.

$$\Rightarrow \underline{e^{-\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2)}\hat{A}e^{\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2)}} = \sum_{n=0}^{\infty}\frac{1}{n!}[-\tfrac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{A}]_{(n)}$$
$$= \hat{A}\left(1 + \frac{\lambda^2}{2!} + \frac{\lambda^4}{4!} + \frac{\lambda^6}{6!}\dots\right) + \hat{B}\left(\lambda + \frac{\lambda^3}{3!} + \frac{\lambda^5}{5!}\dots\right) = \underline{\hat{A}\cosh\lambda + \hat{B}\sinh\lambda}$$

(iv) $[-\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{B}]_{(0)} = \hat{B}$; $[-\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{B}]_{(1)} = \lambda\hat{A}$; $[-\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2),\hat{B}]_{(2)} = \lambda^2\hat{B}$; etc. (roles of $\hat{A},\hat{B}$ interchanged)

$$\Rightarrow \underline{e^{-\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2)}\hat{B}e^{\frac{\lambda}{2}(\hat{B}^2-\hat{A}^2)}} = \hat{B}\left(1 + \frac{\lambda^2}{2!} + \frac{\lambda^4}{4!}\dots\right) + \hat{A}\left(\lambda + \frac{\lambda^3}{3!} + \dots\right) = \underline{\hat{B}\cosh\lambda + \hat{A}\sinh\lambda}$$

Problem 5

(a) $\langle x|p\rangle = \dfrac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}$ (Shankar, p. 137)

(b) $$\underset{\sim\sim\sim}{\psi_{p,\Delta p}(x)} = \frac{1}{\sqrt{\Delta p}}\int_p^{p+\Delta p}\langle x|p'\rangle\,dp' = \frac{1}{\sqrt{2\pi\hbar\Delta p}}\int_p^{p+\Delta p} e^{ip'x/\hbar}\,dp'$$

$$= \frac{\hbar}{\sqrt{2\pi\hbar\Delta p}\; ix}\left[e^{i(p+\Delta p)x/\hbar} - e^{ipx/\hbar}\right]$$

$$= \frac{\hbar e^{i(p+\frac{\Delta p}{2})x/\hbar}}{\sqrt{2\pi\hbar\Delta p}\;\; ix}\left[e^{i\Delta p\cdot x/2\hbar} - e^{-i\Delta p\cdot x/2\hbar}\right]$$

$$= 2\hbar e^{i(p+\frac{\Delta p}{2})x/\hbar}\,\frac{\sin\left(\frac{x\Delta p}{2\hbar}\right)}{x\sqrt{2\pi\hbar\Delta p}} = \frac{e^{i(p+\frac{\Delta p}{2})x/\hbar}}{\sqrt{2\pi\hbar}}\sqrt{\Delta p}\,\frac{\sin\left(\frac{x\Delta p}{2\hbar}\right)}{\left(\frac{x\Delta p}{2\hbar}\right)}$$

(c) $$\lim_{\Delta p\to 0}\frac{1}{\sqrt{\Delta p}}\psi_{p,\Delta p}(x) = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}\underbrace{\lim_{\Delta p\to 0}\frac{\sin\left(\frac{x\Delta p}{2\hbar}\right)}{\left(\frac{x\Delta p}{2\hbar}\right)}}_{=1} = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}} = \langle x|p\rangle$$

(d) $$\int_{-\infty}^{\infty}dx\,|\psi_{p,\Delta p}(x)|^2 = \frac{1}{2\pi\hbar}\Delta p\int_{-\infty}^{\infty}dx\,\frac{\sin^2\left(\frac{x\Delta p}{2\hbar}\right)}{\left(\frac{x\Delta p}{2\hbar}\right)^2} = \quad \left(t = \frac{x\Delta p}{2\hbar}\right)$$

$$= \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sin^2 t}{t^2}\,dt = 1 \quad \text{(HW3)}$$

The wave packets $|\psi_{p,\Delta p}\rangle$ are normalized to 1.

(e) Since $\langle p|p'\rangle = 0$ for $p\neq p'$, $\langle\psi_{p,\Delta p}|\psi_{p',\Delta p}\rangle = 0$

if $|p-p'| = \Delta p$:

$$\langle\psi_{p,\Delta p}|\psi_{p',\Delta p}\rangle = \int_p^{p+\Delta p}dp_1\int_{p'}^{p'+\Delta p}dp_2\int_{-\infty}^{\infty}dx\,\langle p_1|x\rangle\langle x|p_2\rangle =$$

$$= \int_p^{p+\Delta p}dp_1\int_{p'}^{p'+\Delta p}dp_2\,\langle p_1|p_2\rangle = \int_p^{p+\Delta p}dp_1\int_{p\pm\Delta p}^{p\pm\Delta p+\Delta p}dp_2\,\langle p_1|p_2\rangle = 0$$

since $p_1\neq p_2$ everywhere in the integration range.

(f) If I subdivide the p-axis into intervals of length Δp and represent the continuum of vectors $|p\rangle$ in each interval by their "average" $|\psi_{p,\Delta p}\rangle$, these average states are orthonormalized. They are wavepackets of width $\Delta x \sim \frac{1}{\Delta p}$, so they are (contrary to $\langle x|p\rangle \sim e^{ipx/\hbar}$) localized to a finite x-region. For $\Delta p \to 0$, they become larger and larger in x-space, but remain normalizable as long as $\Delta p \neq 0$. Only in the limit $\Delta p \to 0$ normalizability becomes a problem.